CariSurg MedTech Pathways Research Project

Preliminary Proposal

Project Title: Enhancing Emergency Department Triage with Artificial Intelligence

Student Name: Imani Zakuri

Statement of the Problem:

Mercer General Hospital's nurse-led emergency department manages between 70 and 95 patients daily, with volumes exceeding 150 patients during surge periods, placing significant pressure on triage decisions and patient prioritisation. Recent studies have shown that AI-assisted triage models can achieve classification accuracies of up to 80% and improve patient prioritisation (1–3). However, these models have largely been developed using single-site, non-Caribbean datasets and have not been evaluated within Caribbean emergency department workflows. We propose a 12-week pilot to assess whether an AI-assisted decision-support system can improve triage consistency using de-identified Mercer emergency department data.

Previous Work:

Research Paper 1: Use of Artificial Intelligence in Triage in Hospital Emergency Departments: A Scoping Review (1)

This research paper addresses the knowledge gap pertaining to the use of artificial intelligence(AI) and machine learning (ML) in emergency department triage and examines how they affect clinical outcomes. The authors conducted a scoping review to examine the existing use cases of AI in triage by performing a systemized search of the EMBASE, Ovid MEDLINE and Web of Science databases which identified 29 studies published between 2013 and 2023. Overall, the findings showed that ML models consistently outperformed in risk assessment, disease identification, resource allocation, quality of patient care and hospitalization prediction compared to traditional triage methods. While no single AI model was universally superior across all applications, Random Forest, XGBoost and Gradient Boosting models consistently demonstrated the highest predictive performance. The review also considered the ethical issues relating to data and FDA approvals and was limited by its search criteria which excluded papers based on quality appraisal, language and region limiting its overall comprehensiveness and generalizability.

Research Paper 2: TriageIntelli: AI-Assisted Multimodal Triage System for Health Centers (2)

Emergency department overcrowding, increasing case complexity and aging populations have put a strain on healthcare resources which warrants the need for more efficient triage systems. This study aims to explore the integration of AI in triaging systems by evaluating several AI models using data from 1267 patients admitted to two emergency departments from October 2016 to September 2017. Their predictions, based on the Korean Triage Acuity Scale (KTAS) were compared against assessments made by three triage experts. The results showed that the Support Vector Machine (SVM) and Gradient Boosting Machine (GBM) models exhibited the highest prediction accuracies of 78.7% and 79% respectively while a stacked model combining SVM, GBM and Logistic Regression (LR) produces the best overall performance. A key limitation of this study was its reliance on a single dataset which may limit its generalizability across all clinical contexts.

Research Paper 3: Predicting Emergency Severity Index (ESI) level, hospital admission and admitting ward in an emergency department using data-driven machine learning(3)

Emergency departments often face challenges in accurately prioritising patients and allocating resources due to increasing patient volumes and case complexity. This research paper aims to address this problem by developing the XGBoost machine learning model using data from 653,546 emergency department visits collected over 6 years at Mater Dei Hospital in Malta to predict Emergency Severity Index (ESI) level, hospital admission and admitting ward. The model achieved accuracies of approximately 76% for ESI prediction, 83% for admission prediction and 86% for ward prediction, demonstrating the potential of machine learning to support triage and operational decision-making. The study was however limited by its use of a single hospital dataset, limiting the generalisability of the findings to other patient populations.

Research Paper 4: AI-driven Triage in Emergency Departments: A Review of Benefits, Challenges and Future Directions (4)

Traditional triage systems in emergency departments face increasing challenges due to overcrowding, inconsistent patient prioritisation and resource constraints. This study analysed research papers published between 2015 and 2024 sourced through PubMed, Scopus, IEEE XPlore and Google Scholar in order to examine the benefits and limitations of AI-driven triage systems. The findings suggest that AI-driven triage systems can improve patient prioritisation through consistency and objectivity, support real-time decision making through analytics, enhance resource allocation and are adaptable in mass casualty events. The study also identified future opportunities such as algorithm refinement for all clinical contexts, integration with wearable technology and policy development. However, the review is limited by its reliance on existing literature and its limited study selection criteria which may affect its generalizability.

Research Paper 5: Qualitative Exploration of Patient Flow in a Caribbean Emergency Department(5)

This study investigated the organisational factors affecting patient flow in a Caribbean emergency department, addressing the need to better understand barriers and facilitators within developing healthcare systems, The researchers used a qualitative approach involving non-patient observations, observational process mapping and informal interview collecting 203 hours of observations and mapping 143 patient journeys in a major emergency department in Trinidad and Tobago. The study found that patient flow was influenced by six key factors: organisational work processes, department design and layout, material resources, nursing staff levels and roles, non-clinical staff and external departments, with nursing shortages and lack of inpatient beds identified as major barriers. The authors also noted that the study conducted that its use of a single site with one observer limited its transferability and generalisability of the findings to other healthcare settings.

Research Paper 6: A Human-Centered Hybrid AI Framework for Optimising Emergency Triage in Resource-Constrained Settings (6)

Traditional emergency triage systems are often affected by clinician workload, cognitive biases and resource limitations, which can lead to inconsistent patient prioritisation and delayed treatment. This study proposed a human-centred hybrid AI framework that combines cognitive modelling, explainable machine learning, resource-aware optimisation and a human-in-the-loop interface to support emergency triage decision-making. Through simulated and pilot implementations, the framework achieved approximately 90% triage accuracy, reduced false negatives from 12% to 5%, decreased clinician decision-making time by 31%, and improved critical resource utilisation by 20%. The system also allowed clinicians to override AI recommendations and provide feedback, supporting transparency and continuous learning. A key limitation identified by the authors was the small dataset used for development and testing, which may limit the generalisability of the findings across different healthcare settings.

Research Paper 7: LoRaWAN for Smart Medical Emergency Service (EMS) Integrating with AI(7)

This paper addresses delays in emergency medical response, particularly in rural and low-connectivity areas where patients may not receive timely care during critical emergencies. The authors conducted a survey of existing research on LoRaWAN- enabled healthcare systems and proposed a conceptual framework called SmartMed-LoRaWAN, which combines wearable health sensors, LoRaWAN communication, AI-driven anomaly detection, patient triage and telemedicine support. The proposed framework aims to improve response times and patient outcomes by enabling continuous patient monitoring, emergency detection, automated alerts and remote medical guidance. The study identified several challenges including limited bandwidth, data security concerns and scalability issues when supporting multiple patients simultaneously. Since the system has not yet undergone large-scale clinical validation, its real-world effectiveness therefore remains uncertain.

References

1. Tyler S, Olis M, Aust N, Patel L, Simon L, Triantafyllidis C, et al. Use of Artificial Intelligence in Triage in Hospital Emergency Departments: A Scoping Review. Cureus. 2024 May 8. doi:10.7759/cureus.59906

2. Araouchi Z, Adda M. TriageIntelli: AI-Assisted Multimodal Triage System for Health Centers. Procedia Comput Sci. 2024;251:430–7. doi:10.1016/j.procs.2024.11.130

3. Agius S, Cassar V, Magri C, Khan W, Obe DAJ, Caruana G, et al. Predicting Emergency Severity Index (ESI) level, hospital admission, and admitting ward in an emergency department using data-driven machine learning. BMC Med Inform Decis Mak. 2025 Jul 28;25(1):281. doi:10.1186/s12911-025-02941-9

4. Da'Costa A, Teke J, Origbo JE, Osonuga A, Egbon E, Olawade DB. AI-driven triage in emergency departments: A review of benefits, challenges, and future directions. Int J Med Inf. 2025 May;197:105838. doi:10.1016/j.ijmedinf.2025.105838

5. De Freitas L, Goodacre S, O'Hara R, Thokala P, Hariharan S. Qualitative exploration of patient flow in a Caribbean emergency department. BMJ Open. 2020 Dec;10(12):e041422. doi:10.1136/bmjopen-2020-041422

6. Bhuvaneswari E, Prasad KDV, Ashraf M, Jadhav S, Rao TRK, Sudha Rani T. A human-centered hybrid AI framework for optimizing emergency triage in resource-constrained settings. Intell-Based Med. 2025;12:100311. doi:10.1016/j.ibmed.2025.100311

7. Prabhu RV, Balamurugan S, Pradheep R, Karthikeyan K. LoRaWAN for Smart Medical Emergency Service (EMS) Integrating with AI. In: Kalpana R, Sheeba JI, Saruladha K, Akila V, Sathiyamurthy K, Jayabharathy J, et al., editors. Proceedings of the International Conference on Artificial Intelligence and Secure Data Analytics (ICAISDA 2025) [Internet]. Dordrecht: Atlantis Press International BV; 2026 [cited 2026 Jun 13]. p. 1116–27. (Advances in Intelligent Systems Research). Available from: https://www.atlantis-press.com/doi/10.2991/978-94-6239-616-6_81 doi:10.2991/978-94-6239-616-6_81